🧪 **Section: Our Dynamic Universe**

🔷 **Topic: Measure**

Question 16 – Hooke's Law and Elastic Potential Energy

(a) (i) Draw a graph of force F against extension e

✔ 3 marks

✅ **Instructions for the graph (not shown here):**

- **x-axis**: extension e in **mm**
- **y-axis**: force F in **N**
- Scale must be **linear** and **include the origin**
- Plot the points:
  - (3, 1.0), (8, 2.0), (11, 2.9), (15, 3.9), (20, 4.9), (24, 5.9)
- Draw a **best-fit straight line**

(a) (ii) Use your graph to find the spring constant k

✔ 2 marks

From best-fit line:
Choose 2 points (e.g. from data):

- Point A: $(e = 24\,\text{mm}, F = 5.9\,\text{N})$
- Point B: $(e = 3\,\text{mm}, F = 1.0\,\text{N})$

Convert to metres:
$\Delta e = (24 - 3) \times 10^{-3} = 0.021\,\text{m} \quad \Delta F = 5.9 - 1.0 = 4.9\,\text{N}$

$$k = \frac{\Delta F}{\Delta e} = \frac{4.9}{0.021} = 233\,\text{N/m}$$

(a) (iii) Suggest one improvement to increase accuracy

✔ 1 mark

✅ **Answer:**

> Use a **greater range of masses**
> or **repeat measurements and calculate the mean**

Do not accept: "Take more measurements" on its own.

(b) Calculate the elastic potential energy stored at 22 mm extension

✔ 2 marks

Use:
$$E_e = \frac{1}{2}ke^2 \quad k = 240\,\text{N/m (from graph)}, \quad e = 22\,\text{mm} = 0.022\,\text{m}$$

$$E_e = \frac{1}{2} \times 240 \times (0.022)^2 = 5.8 \times 10^{-2}\,\text{J}$$

🧠 Revision Tips

- Hooke's Law: $F = ke$
- Gradient of F vs e graph = **spring constant** k
- **Elastic potential energy** = area under the graph = $\frac{1}{2}ke^2$
